RIMANSTECH INDUSTRIES | ENGINEERING ARCHITECTURE DOSSIER

YOUNG AI EXPLORERS PLATFORM

System Architecture, Generative AI Curriculum Engine & Multi-Tenant EdTech Specification

DOC ID: YAIE-ARCH-2026-V3.4
Principal Architect: Bassey Riman
Stack: Next.js 16.2 / React 19 / Supabase
Status: Production Live (v1.4 Release)

EXECUTIVE ENGINEERING SUMMARY

Young AI Explorers (youngaiexplorers.net) is a multi-tenant, AI-native educational platform and interactive companion to the published Amazon book 'Young AI Explorers' authored by Bassey Riman. Engineered on Next.js 16.2.9 (App Router, Turbopack) and React 19.2.4, the platform combines a structured 38-lesson core curriculum across 6 thematic islands with Vision Vee -- a role-aware generative AI engine powered by Groq (Llama-3.3-70B / 3.1-8B) and OpenAI (gpt-image-1) that synthesizes unlimited custom lessons, validated 3-question quizzes, and custom illustrations with parent/teacher governance.

PRODUCTION TELEMETRY & BENCHMARKS

Live Production:	`https://youngaiexplorers.net`
GitHub Repo:	`basseyriman/Young-AI-Explorers`
Core Curriculum:	`38+ Lessons / 114+ Quiz Items`
Generative AI:	`Unlimited (Llama-3.3 + DALL-E)`
Workbook Engine:	`13-Page Interactive & Print Pack`
Multi-Tenancy:	`4 Portals (Student/Parent/Tchr/Adm)`
Localization:	`5 Locales (EN, FR, ES, SW, AR RTL)`

01 MULTI-TIER SYSTEM ARCHITECTURE & END-TO-END REQUEST TOPOLOGY | Next.js 16 Edge + AI Orchestration + Supabase RLS

TIER 1: CLIENT PRESENTATION & EDGE MIDDLEWARE LAYER (React 19.2.4 + Next.js 16.2.9 App Router)

4 Role-Based Portals
Student, Parent, Teacher & Admin Dashboards (SSR/RSC)

Interactive Learning UI
6 Explorer Islands, 13-Page Activity Pack & Match Quiz

Edge Session Guard
src/middleware.ts
Cookie Auth + Email Gate

5-Locale i18n + RTL
EN, FR, ES, SW, AR (RTL)
Static Dict + Cookie Hydration

`Server Actions` | `POST /api/chat` | `requireRole()` | `Cookie Locale`

TIER 2: SERVER ACTIONS, RBAC ACCESS GATES & VISION VEE AI ORCHESTRATION LAYER

Vision Vee Chat SSE
/api/chat/route.ts
Groq llama-3.1-8b-instant
Role-Aware System Prompt

Topic & Quiz Synthesizer
generate-topic-content.ts
Groq llama-3.3-70b-versatile
JSON Parser + 5-Locale Fallback

Book Illustration Engine
generate-topic-illustration.ts
OpenAI gpt-image-1 (1024px)
+ Branded SVG Fallback

Curriculum & Family Sync
src/lib/db/platform.ts
Parent Approval Gate &
Classroom Roster Engine

`Supabase JWT` | `JSONB Upsert` | `Admin Upload` | `RLS Queries`

TIER 3: MULTI-TENANT PERSISTENCE, ROW-LEVEL SECURITY (RLS) & OBJECT STORAGE (Supabase Cloud)

Identity & Profiles
auth.users + profiles
family_links (Parent->Child)
Role & Consent Metadata

Dynamic Curriculum DB
curriculum_settings &
custom_topics (JSONB)
pending_parent -> ready

Progress & Classrooms
user_badges, quiz_attempts,
classrooms & members
Leaderboard Aggregations

Supabase Buckets & Views
topic-illustrations & free-
resources Buckets + Country
Trending SQL Views

02 LIVE PRODUCTION APPLICATION PARITY & CORE ENGINEERING INNOVATIONS | Verified Live UI Specimens

Fig 1.1: Live Production Landing Hero, Telemetry Counters & Vision Vee Companion

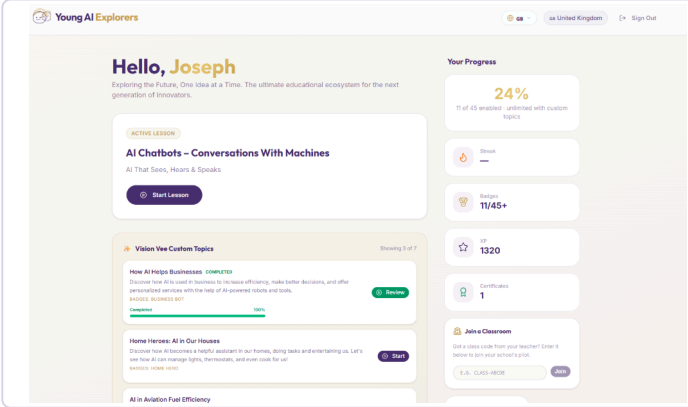

Fig 1.2: Authenticated Student Portal -- Custom AI Topics, XP (1320) & Class Join

CORE ARCHITECTURAL & PRODUCT ENGINEERING INNOVATIONS

1. Hybrid Book-to-AI Curriculum Architecture: Unlike static EdTech courses or unguided chatbot wrappers, Young AI Explorers anchors learning in 38 structured lessons from Bassey Riman's published book across 6 thematic islands (AI Foundations, AI That Sees/Hears/Speaks, Robotics, Creativity, Smart World, Future Worlds), then extends infinitely via Vision Vee.
2. Multi-Model Generative Lesson & Illustration Pipeline: Custom topic synthesis orchestrates Groq (llama-3.3-70b-versatile) for book-style storytelling and 3-question quiz generation alongside OpenAI (gpt-image-1) for 1024x1024 chapter illustrations, backed by deterministic 5-locale fallbacks and dynamic SVG artwork so lesson generation never fails.
3. Child-Safe Parental & Educator Governance State Machine: Student-requested custom topics enter a 'pending_parent' state in PostgreSQL until approved by a linked parent (family_links) or teacher (classroom_members), ensuring strict compliance with the UK Age Appropriate Design Code and UNESCO AI Competency Framework.
4. Dual-Modal Interactive & Printable Activity Pack Engine: A 13-page digital workbook (/activity-pack) featuring an interactive 15x15 algorithmic word search, crossword state validator, robot blueprint studio, and @media print CSS isolation for one-click physical classroom booklet printing.

03 VISION VEE GENERATIVE CURRICULUM DAG & GOVERNANCE STATE MACHINE | Async Synthesis & Formal Invariants

STAGE 1: PROMPT
User / Chat Trigger
Student, Parent, or Teacher submits topic via Chat or Dashboard

→

STAGE 2: GATE
Approval State Machine
Parent/Tchr: 'approved'
Student: 'pending_parent' until Guardian approves

→

STAGE 3: LLM
Groq Llama-3.3-70B
generateTopicContent()
JSON Story + 3 Quiz Qs + 5-Locale Fallback

→

STAGE 4: VISION
OpenAI gpt-image-1
generateTopicIllustration()
1024x1024 PNG -> Bucket + Branded SVG Fallback

→

STAGE 5: READY
Curriculum Activation
content_status='ready'
Unlocks Lesson, Quiz, Match Quiz & Badge

FORMAL CURRICULUM ACCESS, MERGE & QUIZ VALIDATION INVARIANTS (DETERMINISTIC EXECUTION GUARANTEES)

```
I1 (Curriculum Merge):   C_eff(u) = (T_base \ T_disabled(u)) U { t in T_custom(u) | status(t)=='approved' & content(t)=='ready' }
I2 (Student Route Gate): role(u)=='student' => [ access(u, t)==TRUE <=> t in C_eff(u); else HTTP 307 -> /dashboard/student?blocked=t ]
I3 (Quiz Schema Bound):  |Q_valid(t)| == 3  where  forall q in Q_valid(t): |q.options| >= 2 AND q.answer in q.options (else fallbackQuiz)
I4 (Privacy Telemetry):  recordPublicActivity(u)==TRUE <=> profile(u).sharing_level in {'region','global'} (excludes 'private' children)
I5 (Word Search Check):  match(S, W)==TRUE <=> concat(G[r_i,c_i])==W OR reverse(concat(G[r_i,c_i]))==W for W in W_target (15x15 Grid)
```

04 CORE TECHNICAL ALGORITHMS, RESILIENCE FALLBACKS & SECURITY GATES | Codebase-Level Implementation Analysis

A. GENERATIVE AI SYNTHESIS & FAULT-TOLERANT FALLBACKS

1. Structured Lesson & Quiz Synthesis (src/lib/vision-vee/generate-topic-content.ts):
- Invokes Groq's llama-3.3-70b-versatile (temp 0.65) with BOOK_STORY_STYLE_GUIDE and 3 canonical narrative exemplars to match the voice of Bassey Riman's published book.
- Fence-aware JSON extractor extractJsonObject(raw) strips markdown wrappers and isolates the root JSON payload.
- normalizeQuiz() validates each generated question to guarantee >= 2 options and exact answer inclusion. If LLM output is malformed or API keys are absent, fallbackLesson() and fallbackQuiz() synthesize deterministic localized content across all 5 supported languages (EN, FR, ES, SW, AR).

2. Dual-Stage Illustration Pipeline (src/lib/vision-vee/generate-topic-illustration.ts):
- Calls OpenAI Images API (gpt-image-1, 1024x1024) using character consistency prompts (Maya, Alex, Vision Vee).
- Decodes Base64 buffers and uploads via service-role client (src/utils/supabase/admin.ts) to Supabase Storage bucket 'topic-illustrations/custom-topics/{id}.png'.
- If offline or rate-limited, buildTopicIllustrationPlaceholder() generates an inline purple/gold SVG data URI.

B. MULTI-TENANT RBAC, FAMILY SYNC & STREAMING CHAT

1. Role-Aware Streaming Assistant (src/app/api/chat/route.ts & system-prompt.ts):
- Authenticates user session via Supabase SSR and resolves DashboardRole ('student' | 'parent' | 'teacher' | 'admin' | 'guest').
- Streams responses via Vercel AI SDK streamText() using Groq llama-3.1-8b-instant (maxDuration = 30s).
- Dynamically adapts persona rules: never addresses a student as 'your child', and appends structured ---TOPIC--- blocks for one-click curriculum insertion.

2. Guardian Governance & Curriculum Access (src/lib/curriculum-access.ts & platform.ts):
- parentCanManageChild() verifies authority across both family_links and curriculum_settings.managed_by tables.
- syncCurriculumToChild() propagates disabledTopics, sharingLevel, and allowMatchQuiz to linked child accounts.
- requireTopicAccess() enforces server-side redirects if a student attempts to access a locked chapter or an unapproved/pending custom topic, while permitting staff roles ('teacher', 'admin') full preview access.

05 SYSTEM SUBSYSTEM SPECIFICATION MATRIX & LIVE CURRICULUM SPECIMENS | End-to-End Engineering Parameters

SUBSYSTEM MODULE	PRIMARY CODEBASE PATH	CORE ENGINE / PROTOCOL	ARCHITECTURAL INVARIANT & SLA
Edge Auth & SSR Guard	`src/utils/supabase/middleware.ts`	@supabase/ssr Cookie + JWT	Blocks unverified students on protected routes; 0-flash SSR
Vision Vee Chat API	`src/app/api/chat/route.ts`	Groq llama-3.1-8b-instant (SSE)	Role-specific system prompt; sub-400ms TTFT; 30s max duration
Topic & Quiz Engine	`src/lib/vision-vee/generate-*`	Groq llama-3.3-70b + gpt-image-1	Strict 3-question schema check + 5-locale deterministic fallback
Curriculum Access Gate	`src/lib/curriculum-access.ts`	Hybrid DB + Metadata Merge	Enforces parent/teacher topic locks & 'ready' content_status
Classroom & Roster RPC	`src/app/dashboard/teacher/*`	Class Code Join + Leaderboard	Aggregates student badges, quiz avg scores & active topics
13-Page Activity Pack	`src/app/activity-pack/page.tsx`	React State + 15x15 Grid + Print	Interactive word search/crossword + @media print booklet CSS

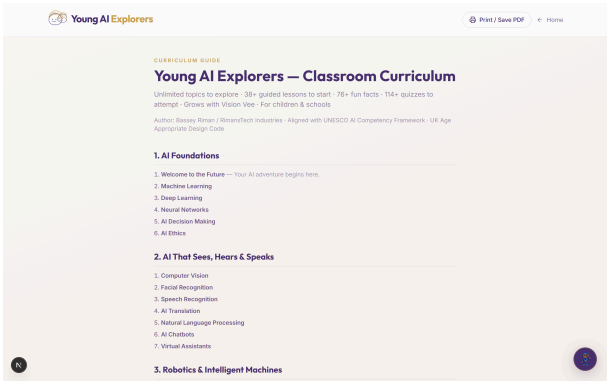

Fig 2.1: Classroom Curriculum Guide (UNESCO & UK AADC Aligned)

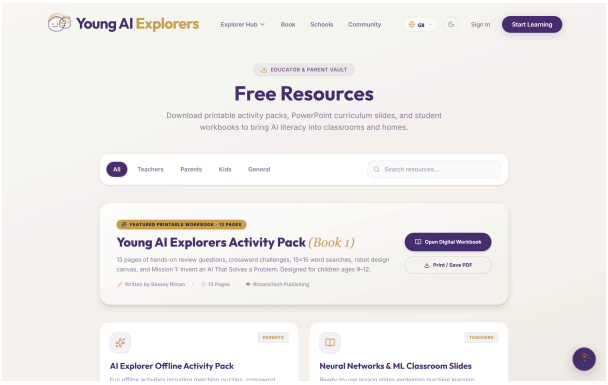

Fig 2.2: Educator & Parent Vault (/free-resources) + Featured Book 1 Pack

06-07 INTERACTIVE ACTIVITY PACK WORKBOOK, MOBILE PARITY & 5-LOCALE I18N ENGINE | Cross-Device & Print Subsystems

SUBSYSTEM ARCHITECTURE SPECS

1. 13-Page Digital & Print Workbook (/activity-pack):
- Built as an interactive stateful React component modelling an authentic educational publisher workbook (Pages 1-13).
- Includes Multiple Choice, True/False, Inline Fill-in-the-Blanks, Memory Matching Dropdowns, 5-Row Home AI Audit, Dream Robot Blueprint Studio, 15x15 Word Search Grid (12 AI terms with forward/reverse coordinate matching), Crossword Challenge, and Mission 1 AI Invention Planner.
- Injects @media print CSS rules to strip navigation chrome and format clean A4 physical worksheets on demand.

2. 5-Locale i18n & RTL Layout Engine (src/lib/i18n/):
- Supports English (en), French (fr), Spanish (es), Swahili (sw), and Arabic (ar).
- Dynamically switches <html dir='rtl'> for Arabic and merges localized lesson overrides with English fallbacks.

3. Global Explorer Communities (/community):
- Aggregates privacy-safe country leaderboards and trending AI topics across 195+ ISO country codes.

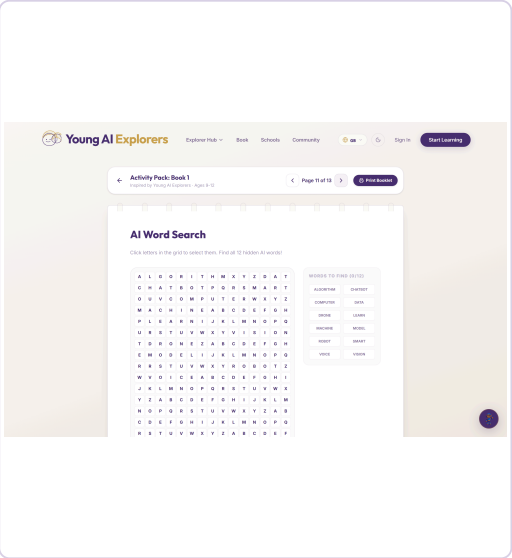

Fig 3.1: Activity Pack 15x15 Word Search (p.11)

Fig 3.2: Mobile Hero

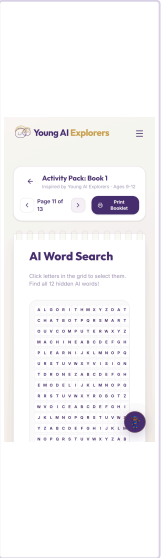

Fig 3.3: Mobile Grid

08 ZERO-TRUST MIDDLEWARE SECURITY & PRODUCTION REPOSITORY TOPOLOGY | Verified Source Tree & Access Controls

PRODUCTION REPOSITORY TOPOLOGY (Young-AI-Explorers/)

```
Young-AI-Explorers/                  # Next.js 16.2.9 + TypeScript 5 Root
|-- src/middleware.ts                # Edge request interceptor & auth guard
|-- src/app/                         # App Router (28 Static & Dynamic Routes)
|   |-- page.tsx & layout.tsx        # Root RTL/LTR i18n + Theme hydration
|   |-- activity-pack/page.tsx       # 13-page interactive & printable workbook
|   |-- api/chat/route.ts            # Groq Llama-3.1-8B SSE streaming endpoint
|   |-- api/community/{countries,trending}/ # Live regional telemetry APIs
|   |-- certificate/[topicId]/       # Verifiable Explorer Certificate generator
|   |-- dashboard/student/page.tsx   # Student learning map, XP & badges portal
|   |-- dashboard/parent/            # Guardian approval gate & child sync RPCs
|   |-- dashboard/teacher/           # Classroom roster, class codes & locks
|   |-- dashboard/admin/             # Platform analytics & blog CMS portal
|   |-- free-resources/page.tsx      # Educator & Parent downloadable vault
|   |-- lesson/[id]/ & quiz/[id]/    # Dynamic book & custom AI lesson/quiz UI
|   |-- match-quiz/page.tsx          # Dynamic memory match game across topics
|   +-- school/{curriculum,demo,pilot,workshop}/ # Institutional B2B portals
|-- src/lib/                         # Core Domain & AI Orchestration Engines
|   |-- auth/dashboard-access.ts     # requireAuth() & requireRole() RBAC guards
|   |-- curriculum-access.ts         # requireTopicAccess() & effective merge
|   |-- db/platform.ts               # 1,014-line Supabase multi-tenant DAL
|   |-- i18n/{i18n.ts,dicts/,lessons/} # 5-locale (EN,FR,ES,SW,AR) engine
|   +-- vision-vee/                  # Generative AI Pipeline
|       |-- generate-topic-content.ts      # Groq Llama-3.3-70B synthesizer
|       |-- generate-topic-illustration.ts # OpenAI gpt-image-1 + SVG fallback
|       +-- system-prompt.ts & book-story-style.ts # Persona & style guides
+-- src/utils/supabase/{server,client,admin,middleware}.ts # SSR & Admin clients
```

SECURITY, PRIVACY & COMPLIANCE GATES

1. Edge Session & Email Verification Gate:
- src/utils/supabase/middleware.ts refreshes HTTP-only auth cookies on every request prior to RSC rendering.
- Unauthenticated requests to /dashboard/*, /lesson/*, or /quiz/* redirect immediately to /login.
- Student accounts lacking email_confirmed_at are isolated and redirected to /signup/verify-email.

2. Strict Role-Based Access Control (RBAC):
- requireRole() resolves authoritative roles from profiles.role with user_metadata fallback, preventing cross-role privilege escalation.

3. Child Privacy & Regulatory Alignment:
- Engineered in alignment with the UK Age Appropriate Design Code (Children's Code), GDPR-K, COPPA principles, and UNESCO AI Competency Framework.
- Public telemetry (shouldRecordPublicActivity) strictly filters out profiles set to 'private' sharing.
- Service-role keys (SUPABASE_SERVICE_ROLE_KEY) are confined exclusively to server-only admin modules (src/utils/supabase/admin.ts).

09 CLOUD DEPLOYMENT ECOSYSTEM, LIVE VERIFICATION & COMMERCIAL TRACTION | Production Infrastructure & Ecosystem

LIVE VERIFICATION ENDPOINTS & CLOUD TOPOLOGY

Production Web App:	https://youngaiexplorers.net
GitHub Repository:	https://github.com/basseyriman/Young-AI-Explorers
Activity Pack Engine:	https://youngaiexplorers.net/activity-pack
Educator Resource Vault:	https://youngaiexplorers.net/free-resources
Classroom Curriculum:	https://youngaiexplorers.net/school/curriculum
School Pilot & Demo:	https://youngaiexplorers.net/school/pilot
Global Communities:	https://youngaiexplorers.net/community
Cloud Infrastructure:	Vercel Edge Network + Supabase PostgreSQL/Storage

COMMERCIAL & EDUCATIONAL ECOSYSTEM

- Published Amazon Book Anchor: 'Young AI Explorers: A Kid's Guide to the Future' by Bassey Riman serves as the physical curriculum foundation, bridged directly into the digital platform.
- Institutional School Readiness: Built-in B2B school pilot onboarding (/school/pilot), workshop booking (/school/workshop), interactive teacher classroom codes, and downloadable PPTX/PDF resource vaults.
- End-to-End Product Velocity: Full CI/CD deployment pipeline from GitHub main branch to Vercel production edge.